

Independent Reading Report: Real-Time Safeguard and Uncertainty-Aware Autonomy

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1 Reading Motivation

This reading project aims to understand the research foundations of **safe autonomous systems under uncertainty**, with a particular focus on Operational Design Domain (ODD), robust perception, runtime perception safety assessment, observation-robust decision-making, and risk assessment. The five selected papers form a coherent technical path: first, defining and refining the safe operating boundary of an autonomous system; second, improving perception robustness to expand this boundary; third, monitoring perception reliability at runtime; fourth, learning decision-making policies that remain robust under imperfect observations; and finally, understanding these topics from a broader risk-assessment perspective.

My own background is in embedded robotics, state estimation, sensor fusion, real-time control, and robot learning. In the **Fast Robots** project at Cornell University, I worked with a small autonomous mobile robot using IMU and ToF sensing, Kalman-filter-based state estimation, PID control, mapping, localization, BLE-based logging, and autonomous maneuver execution. These experiences made me interested in how autonomous systems can remain reliable when sensing, estimation, and control are all affected by uncertainty. Therefore, this reading project is intended to connect my prior engineering experience with future research on safe autonomy, runtime safeguard, and uncertainty-aware decision-making.

2 Selected Papers

No.	Paper	Main Topic
1	<i>Acclimatizing the Operational Design Domain for Autonomous Driving Systems</i>	ODD definition, safety contracts, STL-based evaluation
2	<i>Extending Operational Design Domain for Perception Systems Through Robust Learning</i>	Robust perception, semantic perturbation, ODD extension
3	<i>REAL-SAP: Real-Time Evidence Aware Liable Safety Assessment for Perception in Autonomous Driving</i>	Runtime perception reliability, Bayesian safety assessment
4	<i>Towards Safe Autonomous Driving: Decision Making with Observation-Robust Reinforcement Learning</i>	Observation-robust RL, safe decision-making under perturbed observations
5	<i>Risk Assessment in Autonomous Driving: A Comprehensive Survey of Risk Sources, Methodologies, and System Architectures</i>	Risk sources, methodologies, data foundations, system architecture

3 Paper 1: ODD Acclimatization for Autonomous Driving Systems

Core Idea

The first paper studies how to define and refine the **Operational Design Domain** of autonomous driving systems. The key motivation is that learning-based autonomous systems may become unsafe even when their hardware and software components are functioning normally. These risks are closely related to **Safety of the Intended Functionality (SOTIF)**, where functional limitations, domain shift, adverse weather, poor lighting, or unfamiliar scenarios may cause unsafe behavior.

The paper argues that an ODD should not be treated as a static descriptive list. Instead, it should be formalized, tested, and iteratively refined based on closed-loop system behavior. The authors use structured environmental modeling and Signal Temporal Logic (STL) safety contracts to evaluate whether an autonomous driving policy remains safe under different environmental configurations.

A representative lane-keeping safety requirement can be expressed as

$$\Phi_{SL} = \Box \left(d < \frac{w}{2} \right),$$

where d is the lateral distance from the lane center, w is the lane width, and $\Box$ means that the condition should hold over time. The paper also evaluates perception error over temporal windows, emphasizing that safety should not be judged only frame by frame. A small perception error may be acceptable at one instant, but temporally accumulated error can eventually lead to unsafe closed-loop behavior.

Takeaway

My main takeaway is that safe autonomy is a **system-level property**. It is not enough to evaluate perception accuracy, planning quality, or control stability separately. A safe autonomous system needs to know under what operating conditions it can be trusted, and it should refine this boundary through testing and observed failures. This directly connects to my Fast Robots experience: noisy ToF measurements or IMU drift were not only sensor-level issues, because they affected distance control, yaw estimation, localization, and final robot behavior.

4 Paper 2: Extending ODD Through Robust Learning

Core Idea

The second paper studies how to extend the ODD of autonomous driving perception systems through robust learning. While the first paper focuses on defining and evaluating safe operating boundaries, this paper asks how a perception module can become robust enough to safely operate in a broader range of conditions.

The core idea is to model a new operating condition as a **semantic perturbation** of the original data distribution:

$$x' = G(x, \delta),$$

where x is the original input, x' is the perturbed input, G is the perturbation model, and δ is a semantic nuisance parameter. Unlike norm-bounded adversarial perturbations, δ has physical or semantic meaning, such as fog density, brightness level, weather intensity, or simulation-domain style.

The paper evaluates robustness at both data-wise and distribution-wise levels. Some samples remain stable under semantic perturbation, while others become weak or fragile. These weak data points

are useful because they reveal edge cases and can guide targeted retraining, data collection, and continuous engineering. The paper further proposes robust learning methods based on model-based augmentation and worst-case learning:

$$\min_w \mathbb{E}_{(x,y) \sim D} \left[\max_{x' \in \Delta(G, \epsilon)} L(x', y; w) \right].$$

This formulation encourages the perception model to remain reliable under the most damaging semantic variations within the candidate operating domain.

Takeaway

This paper helped me understand that ODD extension is both a safety problem and a learning problem. In real autonomous systems, the most important perturbations are often not artificial pixel-level changes, but physically meaningful domain shifts such as fog, rain, low illumination, reflective surfaces, sensor degradation, or sim-to-real differences. This is relevant to my embedded robotics background: ToF readings can be affected by surface reflectivity, IMU estimation can drift, and visual sensors can be degraded by lighting changes. These are not random disturbances only; they are operating-domain changes that should be explicitly considered in robust autonomy.

5 Paper 3: REAL-SAP for Runtime Perception Safety Assessment

Core Idea

The third paper proposes **REAL-SAP**, a real-time evidence-aware safety assessment framework for learning-based perception systems. Its central question is: even if a perception model performs well offline, how can an autonomous system know whether the current perception output is reliable during deployment?

The paper argues that high-level autonomy requires both **situational awareness** and **self-awareness**. The vehicle must perceive the surrounding environment, but it must also estimate whether its own perception result can be trusted. The proposed framework combines offline testing knowledge and real-time evidence in a Bayesian network to estimate the probability of abnormal perception events:

$$P_F(A|\Omega, \Xi) = G(P_{f_1}(A|\Omega, \Xi), \dots, P_{f_N}(A|\Omega, \Xi)),$$

where A denotes an abnormal event, Ω is offline knowledge, Ξ is real-time evidence, and f_i denotes an individual perception subsystem or modality.

A key strength of REAL-SAP is that it does not rely on one single signal. It integrates ODD information, neural-network confidence, localization, HD-map references, and redundancy-based feature matching. For example, camera-based lane detection can be compared with map-based lane references. If the detected lane features and reference features disagree, the mismatch becomes evidence that the perception result may be unreliable.

Takeaway

The most important insight is that **uncertainty estimation is not the same as safety assessment**. A neural network may output a high confidence score even when the actual perception result is wrong. Safety assessment asks a broader system-level question: given the current operating condition, offline validation knowledge, and real-time evidence, can the perception result be trusted for downstream autonomy?

This paper strongly connects to my experience with Fast Robots. Kalman filtering improved noisy ToF-based distance estimation, but the estimator itself did not explicitly decide whether the current estimate was safe enough for control. A REAL-SAP-like runtime monitor could use onboard logs, sensor consistency, motion-model prediction, and operating conditions to generate reliability signals. If IMU yaw, ToF distance, and motion prediction become inconsistent, the robot could slow down, switch to conservative control, or trigger a fallback behavior.

6 Paper 4: Observation-Robust Reinforcement Learning

Core Idea

The fourth paper moves from perception safety to decision-making robustness. It studies how autonomous driving policies can remain safe when observations are noisy, perturbed, or unreliable. Standard reinforcement learning policies may achieve high reward in simulation, but their actions can change significantly under small observation errors. In safety-critical driving, such sensitivity can cause unsafe lane changes or collisions.

The paper formulates the problem as an **observation-robust Markov Decision Process** and proposes an observation-robust actor-critic method. The objective is to maximize expected return while constraining policy variation under adversarial observation perturbations:

$$\max_{\pi} \mathbb{E} \left[\sum_{t=0}^T \gamma^t r(s_t, a_t) \right], \quad \text{s.t. } \mathbb{E}[c(s, \pi, \Delta)] \leq \epsilon.$$

The policy variation is measured using Jensen–Shannon divergence:

$$c(s, \pi, \Delta) = D_{JS}(\pi(a|s) \parallel \pi(\tilde{a}|s + \Delta)).$$

An adversarial agent is trained to generate strong observation perturbations, while the driving policy learns to remain stable under these perturbations. The method is evaluated on highway lane-change tasks in SUMO and compared with DQN, PPO, and discrete-action SAC baselines.

Takeaway

This paper helped me connect perception uncertainty with downstream decision-making. Even if perception is imperfect, a safe autonomous system should not react too aggressively to small observation changes. Observation-robust RL provides a way to train policies that are less sensitive to noisy inputs.

This is relevant to both my Fast Robots and Overcooked-AI experiences. In Fast Robots, ToF measurements and IMU yaw estimates were imperfect, and aggressive responses to noisy estimates could destabilize control. In Overcooked-AI, policies also needed to handle interaction uncertainty and shared-space constraints. This suggests that robust decision-making should consider not only sensor noise but also uncertainty from other agents’ behavior.

7 Paper 5: Risk Assessment Survey

Core Idea

The fifth paper provides a comprehensive survey of risk assessment in autonomous driving. It is useful as a synthesis because it organizes safe autonomy research from the perspectives of **risk sources**, **risk assessment methodologies**, **data foundations**, and **system architectures**.

The paper classifies risk sources into three main categories: perception uncertainty, prediction uncertainty, and decision-making uncertainty. Perception uncertainty comes from sensor noise, occlusion, poor lighting, rain, fog, or algorithmic limitations. Prediction uncertainty comes from the difficulty of forecasting other agents' future behavior. Decision-making uncertainty arises when the system must choose safe and efficient actions under incomplete information and conflicting objectives.

The survey also classifies risk assessment methods into rule-based methods, probabilistic-statistical methods, equivalent-model-based methods, and deep-learning-based methods. Rule-based methods are interpretable but limited in open environments. Probabilistic methods are useful for uncertainty modeling but depend on distribution assumptions. Equivalent model methods, such as artificial potential fields or risk fields, are intuitive for spatial risk modeling. Deep-learning methods can handle complex scenarios but often require large-scale data and may lack transparency.

A particularly important point is **capability-aware risk assessment**. Risk is not only determined by the external environment. It also depends on the autonomous system's own perception, decision-making, and control capability. The same driving or robotic scenario may be safe for one system but unsafe for another.

Takeaway

This survey helped me understand how the previous four papers fit into a larger research landscape. ODD defines operating boundaries, robust learning expands perception capability, REAL-SAP monitors runtime perception reliability, and observation-robust RL improves decision-making under uncertain observations. Together, these topics can be viewed as components of a broader risk assessment and runtime safeguard framework.

The capability-aware view is especially relevant to my background. In my robot experiments, the same indoor environment could be safe or unsafe depending on robot speed, sensor quality, control gains, localization accuracy, and state-estimation stability. This suggests that risk assessment for autonomous systems should be adaptive to both environmental complexity and system capability.

8 Integrated Understanding Across the Five Papers

After reading these five papers, I understand safe autonomy as a layered problem rather than a single algorithmic task. A reliable autonomous system should be able to answer four connected questions.

First, where can the system operate safely? This is the role of ODD definition and refinement. The system needs a formal, testable boundary for safe operation.

Second, how can this boundary be expanded? Robust learning can improve perception modules under meaningful domain shifts, allowing the system to handle a broader range of operating conditions.

Third, is the current perception result reliable? Runtime safety assessment, such as REAL-SAP, estimates whether the current perception output can be trusted using offline knowledge and real-time evidence.

Fourth, how should the system act when observations are uncertain? Observation-robust decision-making aims to ensure that planning or policy outputs remain stable and safe under perturbed observations.

These four questions can be summarized as the following pipeline:

ODD Definition → Robust Perception → Runtime Safety Assessment → Robust Decision-Making → Safe Control

This pipeline is highly relevant to my research preparation because it connects my practical experience

in embedded robotics with broader research problems in safe autonomous systems. My prior work gave me hands-on experience with noisy sensors, state estimation, real-time control, and failure analysis. These readings help me reinterpret those experiences through the lens of safety, uncertainty, and runtime safeguard.

9 Connection to My Background and Future Direction

My experience in Fast Robots showed that reliable autonomy cannot be achieved by a nominal controller alone. The robot’s behavior depended on a chain of modules: ToF sensing, IMU yaw estimation, Kalman filtering, PID control, state-machine logic, BLE logging, and offline analysis. A failure in one part could propagate to the entire behavior. This matches the survey paper’s view that risk can originate from perception, prediction, decision-making, and control, and that these risks are coupled.

A possible future direction is to develop **runtime safeguard mechanisms for small autonomous robots under sensor uncertainty**. Such a system could monitor the consistency among ToF distance, IMU yaw, motion-model prediction, and control response. If the reliability score drops, the robot could automatically switch to a safer behavior, such as slowing down, re-estimating its state, stopping, or executing a conservative fallback controller.

Another direction is to connect **robust learning and weak-data extraction** with robot logs. In Fast Robots, failure cases were recorded through onboard logs containing time, distance, yaw, target yaw, PWM, and state information. These logs could be used to identify weak operating conditions where sensing or control becomes unreliable. This idea is similar to robust ODD extension and weak data extraction in autonomous driving perception.

A third direction is to study **uncertainty-aware decision-making in interactive environments**. My Overcooked-AI project exposed me to multi-agent coordination under shared-space constraints. Combining this with observation-robust RL suggests a broader research question: how can autonomous systems remain safe when uncertainty comes from both sensors and other agents’ unpredictable behavior?

10 Potential Research Questions

Based on the five papers, I am interested in the following research questions:

1. How can ODD and risk-assessment concepts be adapted from autonomous driving to small-scale mobile robots and general cyber-physical systems?
2. How can semantic perturbation models be designed for robotic sensing, including ToF, IMU, camera, and LiDAR observations under real-world environmental changes?
3. How can runtime perception reliability be estimated using lightweight Bayesian or probabilistic models on embedded autonomous platforms?
4. How can perception reliability scores be propagated to downstream planning, fallback behavior, or conservative control strategies?
5. Can onboard robot logs be used to automatically extract weak or risky operating conditions for safety validation and model improvement?
6. How can observation-robust reinforcement learning be combined with runtime safety monitors or control safeguards to improve decision-making in uncertain and interactive environments?

11 Conclusion

This reading project provided a structured understanding of safe autonomy under uncertainty. The five papers together show that autonomous system safety requires more than accurate perception or high-reward decision-making. It requires formal operating-domain definition, robust perception under semantic domain shifts, runtime perception reliability assessment, robust decision-making under uncertain observations, and system-level risk assessment.

For my future research preparation, the most important insight is that safe autonomy should be treated as an integrated perception–estimation–decision–control problem. My background in embedded robot control, Kalman-filter-based state estimation, sensor fusion, and robot learning provides a practical foundation for this direction. Building on these readings, I am particularly interested in real-time safeguard and capability-aware risk assessment for autonomous robotic systems operating under uncertain, dynamic, and interactive real-world conditions.